

Local Free-View Neural 3D Head Synthesis for Virtual Group Meetings

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Figure 1: These two images show 3D anaglyph representations of new virtual agents. Using cyan-red glasses will show these faces in 3D.

ABSTRACT

Virtual group meetings provide enormous potential for remote group communication. However, today's video conferences incur numerous challenges compared to face-to-face meetings. For instance, perception of correct gaze, deictic relations, or eye-to-eye contact is impeded due to the fact that the camera is offset from the eyes of the other users' avatars and that the gallery view is different for each group member.

In this paper, we describe how 3D neural heads can be synthesized to overcome these limitations. Therefore, we generate different head poses using a generative adversarial network for a given source image frame using state-of-the-art technology. These head poses can then be viewed in a local space to freely control the gaze of the head poses. We introduce and discuss five use cases for these synthesized head poses that aim to improve intelligent agents and virtual avatar representations in regular video group meetings.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models—; Computing methodologies—Artificial intelligence—Distributed artificial intelligence—Intelligent agents

1 INTRODUCTION

Group meetings have become a staple for interaction and exchange of information in many areas, including education and business. Recently, numerous group meetings have transitioned from traditional face-to-face (F2F) meetings to virtual group meetings. Such virtual (or hybrid forms of) group meetings have received considerable attention as a widespread alternative to F2F communication, and we

already observe a transition of many F2F group meetings to virtual forms. They allow multiple persons to communicate remotely, for instance, via video conferences or immersive platforms, and, therefore, provide more sustainable means of group meetings with reduced travel and CO2 emission than F2F meetings. Forced by the COVID-19 pandemic, people around the globe are experiencing virtual and hybrid group meetings as an accessible and sustainable, but also sometimes frustrating way of remote group communication. Furthermore, virtual meetings incur numerous technological limitations, human-computer interaction (HCI) challenges, and psychological barriers. For example, most current virtual meeting solutions hinder perception of correct gaze, body language, deictic relations, or eye-to-eye contact.

Artificial intelligence (AI) has enormous potential to improve virtual meetings such that they can become an equivalent (or even better) alternative to F2F meetings. AI-powered features such as background blur or replacement, noise suppression, and automatic translations and transcriptions have already enhanced the interactive meeting experience. Furthermore, in video conferences, the perception of others or proximity, i.e., the spatial relationships between individuals, is impeded, since each participant perceives an individual arrangement of the other participant's videos.¹

Virtual group meetings typically contain audiovisual representations of group members (e.g., pictures, videos, or 3D models), so-called *avatars*, as well as a virtual space in which the group meets [30]. Such audiovisual immersion is one of the most essential aspects of virtual group meetings and allows users to perceive a sense of spatial and social presence [13, 30] in which users ideally perceive the illusion of being in the same virtual space (i.e., place illusion) [29].

Advancements in AI could further improve meeting experiences,

¹Microsoft Teams as well as Zoom recently addressed this issue by the so-called *Together Mode* and *Immersive View*, which uses CV to segment the upper body of participants to display them all in a shared, consistent virtual space.

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for instance, in the area of artificial social intelligence. In this context, the displayed head of a user's video avatar or intelligent virtual agent could simulate human behavior by gazing towards the speaker or establishing and maintaining eye-to-eye contact.

In this paper, we introduce neural 3D head syntheses for virtual group meetings, which can freely rotate their head. These synthesized head poses can then be viewed in a local space to freely control the gaze of the head poses. The remainder of this paper is structured as follows. Section 2 resumes related work. Section 3 explains the synthesis process of the free-view 3D head poses. In Section 4, we introduce and discuss five use cases for these synthesized head poses, which shows different concepts to improve intelligent agents and virtual avatar representations in regular video group meetings. Section 5 concludes the paper and provides an overview about future work.

2 RELATED WORK

Virtual group meetings: Literature suggests that the technology used in a group meeting affects how we communicate, collaborate, learn, work, and make decisions [6, 25, 37]. Therefore, it is important to understand how various virtual meeting formats and technologies affect the fine-grained behavioral patterns at the core of group meetings. Until now, people often consider virtual meetings an inferior alternative [4, 14], are reluctant to use conference calls or video conferencing, and tend to prefer F2F meetings [16, 20, 23]. Indeed, F2F communication has a number of advantages. For example, when communicating via F2F, a speaker can draw on numerous visual and auditory back channels to receive real-time feedback from the audience, and then rapidly adjust as necessary [31]. In contrast, virtual group meetings have some major advantages. For instance, (differently abled) people around the globe have greater access to virtual than F2F meetings, making it easier to create more diverse and therefore potentially more creative teams [35]. Virtual meetings might also improve creative outputs through integrated documentation and collaboration tools. For instance, usually while one team member talks in F2F meetings, the others have to remain silent, which might hinder team creativity and ideation [7]. Virtual formats can overcome such challenges via chats or breakout-rooms, which, however, could lead to distractions.

Social signals in group meetings: Virtual group meetings, however, incur numerous technological, psychological, and societal challenges [17, 25, 28]. Due to missing or limited (non-)verbal communication cues, these have a profound and potentially negative impact on how we perform business, coordinate our actions, build and maintain relationships, make decisions, and exert social influence through social interactions [1, 17, 37]. Complex multi-modal signals shape interactions in meetings as they involve the “three Vs of communication”: (i) vocal, (ii) visual, and (iii) verbal. This includes spoken words, text, images, as well as non-verbal behavior [32].

After interpreting social cues in F2F conversation, humans provide non-verbal and para linguistic feedback by nodding and smiling, or altering their pitch, volume, and quality of voice when answering [34]. Perceiving an interacting partner's gaze, deictic references, or affective expressions is often not possible in virtual settings, which can have a negative effect on group dynamics. For instance, latency-induced delays of > 1.2 seconds make people perceive their interlocutors as less friendly or less focused [26]. Moreover, the absence of eye-to-eye contact in video conferencing has been shown to cause a lack of trust [3, 19]. This is problematic in light of findings that the association between trust and team effectiveness is stronger in virtual meetings than in F2F meetings [5]. Furthermore, the arrangement of video avatars in the gallery view, which is often used in video conferences with multiple participants, is typically different for each user. Hence, most of these video conferences lack a shared space, eye-to-eye contact, and consistent gaze among participants.

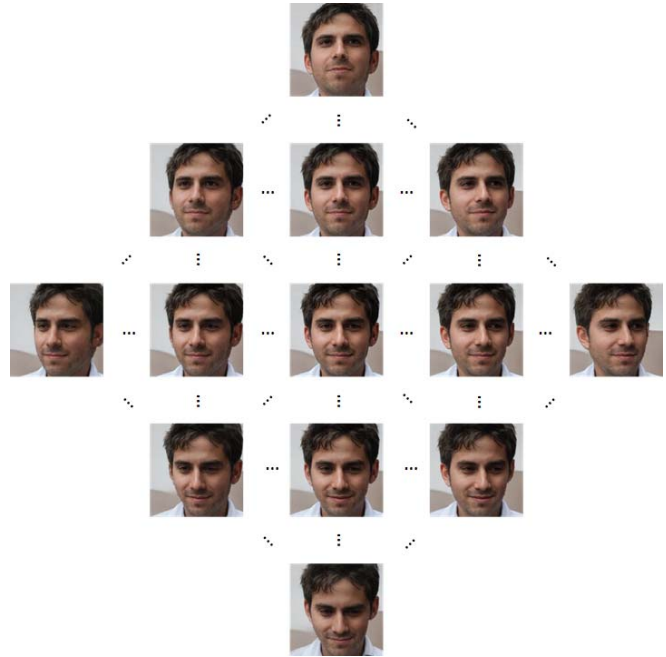


Figure 2: We pre-generated a total of 1861 images for each agent, depicting a pitch and yaw rotation in $[-30, 30]^\circ$.

Artificial intelligence in virtual group meetings: Artificial intelligence (AI) has enormous potential to improve group interaction dynamics and attendee experiences in virtual meetings. Already today, computer vision (CV), natural language processing (NLP), and (social) signal processing support AI-powered features such as background blur or replacement, noise suppression, automatic transcriptions and real-time translation. Another important research area focuses on the challenge to use AI to improve video quality and reduce bandwidth usage. Wang et al. demonstrate such methods [36] and describe that their work can benefit virtual group meetings by simulating F2F experiences. They use generative adversarial networks (GAN) [9] with inputs for roll, pitch, yaw and eye rotation to generate images and videos using only a single driving image as a source for the RGB data. A lot of work has also been done in the area of gaze redirection in regard to editing videos, photos and establishing eye contact in video conferences [11, 12, 21, 27, 39–41].

3 GENERATING FREE-VIEW 3D AGENTS AND AVATAR

Humans have evolved to easily determine if they are being looked at by another person [8]. In a F2F setting, this also makes it easy to get a feel for the room and who everyone is interacting with. It also helps understand the current mood of listeners, and people who are speaking can understand if they hold the listeners' attention. In video group meetings, this does not work as well, since users usually do not look directly into the webcam and the video is only rendered in 2D, thus removing any 3D hints that could be picked up. To improve the cues for socially engaging meetings, we came up with a method using the results of [36] to quickly create 3D representations of realistic avatars, as well as create unlimited new agents on the fly.

To create a new agent, we first extract a face from the publicly available implementation of the StyleGAN2 generator [18] at *TPDNE*². This implementation of the StyleGAN uses a dataset of human faces, and can create completely new images of nonexistent humans. We need to generate a neutral face for the agent, so that

²<https://www.thispersondoesnotexist.com/>

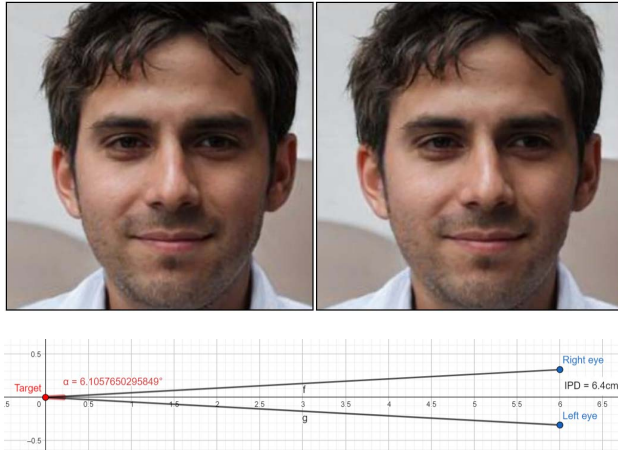


Figure 3: Top row shows 2 images of the same face but with the yaw rotated by 6°. See figure 1 for an anaglyph view of the two images above. The bottom diagram shows an example calculation for the angle which results from an IPD of 64mm and a distance to screen of 60cm

they are inconspicuous in a group video meeting. We can do this by either manual selection or by using a mood determining neural network [2] to evaluate if a generated face is neutral. Once we have extracted a new face, we use the GAN described by [36], to create a grid of novel viewpoints (figure 2) which we will later use to move and animate the agent.

To create a 3D image, we need to generate a different perspective for each eye. The mean average interpupillary distance (IPD) for males is estimated to be around 64 mm [10]. Healthy distances users should keep to their monitors are estimated to be between 51cm and 101cm. We used these values to calculate the necessary angle between two head orientations to facilitate a 3D representation. In figure 3, you can see an exemplary calculation for an IPD of 64 mm and a distance-to-screen of 60cm that results in a 6.1° angle. For [51, 101] cm, the resulting angle is in [7.18, 3.63]°. Since our pre-generated images only show difference of at least 1°, we can currently only select integer values. By crossing eyes, a 3D image should already be visible in figure 3. For use in a video conference, we currently suggest creating a red-cyan image that can be viewed with anaglyph [24] [38] 3D-glasses, as shown in figure 1. We can now use the generated images to create a free-view 3D-representation of a newly generated agent for our group video meetings.

To generate an avatar for a real participant, we simply need to use a single image with a neutral expression, and the same method as described above.

3.1 Ethical Discussion

While working with biometric data such as faces, it is important to consider the ethical implications when using these in any data driven model. When creating avatars, the usage, transmission and storage of any data has to be managed according to current GDPR³ regulations. This is a standard procedure, but when working with data generated by a GAN such as in our case, we have to be aware that data leakage in GAN models can technically occur and its impact is yet inconclusive [33]. At worst, the faces generated could relate to those that are in the training set and be re-identified by those persons. When using these agents on a larger scale, we suggest

³<https://eur-lex.europa.eu/eli/reg/2016/679/oj>

using a similar method as used in [33] by running face matchers on the known training data with any agent that should be used in production. Another ethical issue to be aware of is the introduction of a social bias which depends on the type, amount and quality of training data used [15].

4 USE CASES

In the following use cases, we describe how prototypes of the 3D-generated agents and avatars can potentially benefit virtual and video group meetings.

4.1 Use Case 1: Stereoscopic Representations of Agents & Avatars

The 3D-representations generated using the method described above look very realistic and work without any filters or 3D-mappings. These agents and avatars "pop-out" from your screen and add spatial cues that viewers can experience a more natural F2F meeting as opposed to looking at 2D videos. While the method described above renders the agents at about zero parallax, we can utilize the same parameters to achieve a pop-out effect by reducing parallax, i.e. increasing the angle between the two images shown. This causes the images to "pop-out", which can increase the saliency of that individual. By increasing the parallax, the agent appears as if it were a distance behind the screen, i.e. in the background. Using these two effects gives us interesting possibilities to influence the meetings.

1. To put emphasis on the current speaker, we can apply a pop-out effect on the avatar/agent, to make the speaker more salient and faster to find among the other participants.
2. Using the DPI value of the current screen and size of the participants videos, we can choose the appropriate strength of the effect to simulate placing the 3D representations at an adequate distance behind/before the screen so that the 3D agent/avatar will appear the same size as a regular person.

4.2 Use Case 2: Gaze Dependent Eye Contact

As elaborated by [40], eye contact remains an important method of communication. The authors of [12] also propose a method where they translate participants images in a way, that it appears as if they make eye contact when looking at each other's images in a video group meeting. Similar to [12], we estimate the current gaze point on a screen using the user's webcam with the tool created by [22]. Using this estimate, we can determine where every user is looking at in the conference. Using a web server, we can collect and aggregate this data to render the appropriate agent representation image using the [36] generator. From our pre-rendered images, we pick the image with the rotation closest to the direction of the partners images on screen.

4.3 Use Case 3: Gaze Consistency

As demonstrated in figure 4, we can facilitate gaze consistency throughout all participants of the conference, using the same method as in the previous use case. Avatars with consistent gaze, give the virtual meeting more interaction. Even passive listeners, are now active participants in the meeting in the way of establishing eye contact either with the speaker, or other listeners. All participants can now follow the flow of the meeting more easily by seeing who is being looked at and spoken to. Further, studies will investigate this effect in depth. Generated Agents also could help participants follow the conversation by looking at the current speaker, and vice versa, give the current speaker people they can make eye contact with while speaking.

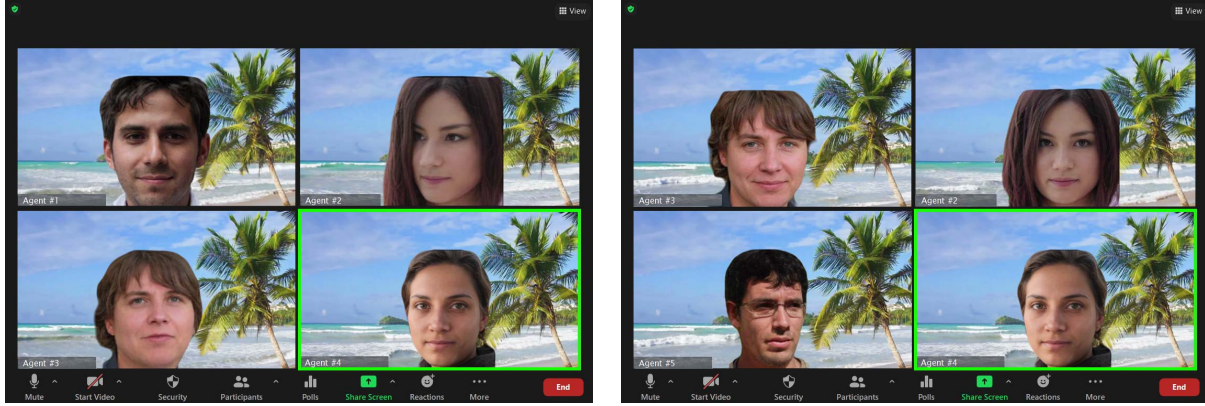


Figure 4: By modifying the gaze directions for each view, we can reflect the current state of a Zoom conference. The figures show a conference from the point of view of two different participants. The first figure shows the perspective of Agent #5 (who is looking at the active speaker #4). We can clearly see, Agent #1 and #4 looking at us, while #2 and #3 are looking at Agent #1. In the second figure, we can see the conference at the same time from the perspective of Agent #1. Thus, Agent #2 and #3 are pictured making eye contact, #5 is looking at #4. The active speaker #4 is always pictured as making eye contact. In each perspective, the gaze between all the participants remains consistent.

4.4 Use Case 4: Directional Audio Consistency

Furthermore, in F2F meetings, you are able to talk to your neighbors quietly during a meeting for quick exchanges. We can connect users and groups of users that are currently looking at each other, and thus, dynamically influence the conference. When these users lock gaze, we can assume they want to talk to each other, and establish an additional audio channel just them. They will still be able to follow the main discussion, but are free to talk to each other briefly, without interrupting the conference. If groups that are established in such a way make up the majority of a conference, the main audio feed could be transferred to this group, making it the main audio channel.

4.5 Use Case 5: Primarily one directional meetings

In large, primarily one directional meetings with limited interaction e.g. lectures and panel discussions, where participant numbers are too large to show in a meaningful way, we can explore the use of agents as a representation of subsets of the participants. I.e. 100 participants could be represented by 10 agents, which in turn each reflect the average or aggregated mood, agreement or disagreement of all participants. The speaker(s) could naturally understand the state of the room at a glance. We can achieve this aggregation by using a mood detecting neural network such as [2], which is able to estimate the current mood of all faces on a single image in real-time. Furthermore, this feature could be extended to represent users of different genders and ethnicities to ensure everyone is being seen.

5 CONCLUSION & FUTURE WORK

In this paper, we have demonstrated our prototype which is able to generate stereoscopic representations of 3D-agents and avatars for the use in video group meetings. We plan to develop a real-time GAN solution, which we can use to further explore and evaluate the use cases described in future studies. The use of quickly generated 3D-agents is especially promising, since they show a realistic human agent and, as passive participants, have the potential to reduce virtual-meeting-fatigue by giving the participants stereoscopic 3D-faces to look at. In future developments, we plan to improve the agents to be more interactive and inhabit a more dominant role in video and virtual group meetings.

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⁴<https://www.nvidia.com/>

⁵<https://thispersondoesnotexist.com/>

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